Shibo Yan

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Education

Southwest University | Software Engineering | Full-time
College of Computer and Information Science & College of Software (the College)
Master's Program Sep 2024 - Jun 2027
Campus Honors: Outstanding Individual in Academic and Technological Innovation; First-Class Academic Scholarship; Outstanding Individual in Campus Cultural Activities, among others.
Bachelor's Degree Sep 2019 - Jun 2023
GPA: 3.95/5.00; rank in major: 8/124.
Course grades: Data Structures 98, Probability Theory 98, Data Mining 94, Software Modeling and Design 96.
Campus Honors: Second-Class Scholarship; Merit Student; Civic Conduct Award; Outstanding Communist Youth League Member, among others.

Research Experience

Southwest University | RISE | Safe Control & Formal Verification Sep 2024 - Jun 2027
Centre for Research and Innovation in Software Engineering | Trustworthy Software Laboratory
Research: learning-aided reach-avoid controller synthesis with formal probability bounds for nonlinear stochastic systems.

- Participate in a National Natural Science Foundation of China General Program project, "Polynomial Algebraic Methods for Safety Verification of Neural-Network-Controlled Systems" (No. 62472362; Jan 2025 - Dec 2028).
- **Conference:** **Shibo Yan**, Xia Zeng, Hengjun Zhao, Zhengfeng Yang, Zhiming Liu. *Learning-Aided Reach-Avoid Controller Synthesis with Formal Guarantees for Stochastic Systems*. MEMOCODE 2026 (ESWEEK 2026). 1st author, published.
- **Journal:** Xia Zeng, **Shibo Yan**, Zhiming Liu, Zhengfeng Yang. *Safety Guarantees for Nonlinear Systems with Bayesian Neural Network Control Policies*. *Formal Aspects of Computing* (CCF-B). 2nd author (supervisor 1st), submitted.
- **Conference:** Xia Zeng, **Shibo Yan**, Hengjun Zhao, Zhiming Liu, Zhengfeng Yang. *Formal Reach-Avoid Controller Synthesis for Stochastic Systems via Iterative Neural-Symbolic Learning*. AAAI (CCF-A, round-1 review passed). 2nd author (supervisor 1st), submitted.
- **Conference:** **Shibo Yan**, Hengjun Zhao, Xia Zeng, Zhengfeng Yang, Zhiming Liu. *Safe Controller Synthesis via Sparse Polynomial Learning and Efficient Barrier Certificate Verification*. DAC 2027 (CCF-A). 1st author, under submission.

Projects and Internship Experience

Larac-Drive | Reinforcement Learning for Vehicle Control and Probabilistic Reach-Avoid Verification under Stochastic Disturbances | Independent Developer Mar 2025 - Present

- **Background:** Studied joint lateral and longitudinal control for adaptive following, hard lead-vehicle braking, and cut-ins on structured roads, balancing tracking, safety, and smoothness under braking uncertainty and vehicle motion constraints.
- **Work:** Modeled no-sideslip vehicle kinematics in Frenet coordinates with longitudinal acceleration and curvature inputs. Trained a Soft Actor-Critic (SAC) policy with a reward that penalized tracking error and control input rates. Implemented IDM and MPC baselines. Used a robust control barrier function (CBF) for time headway and a high-order CBF (HOCBF) for road boundaries. An online quadratic program (QP) enforced both constraints and penalized command changes, with emergency braking at low speeds or when the QP was infeasible. For stochastic braking by a same-lane lead vehicle, modeled a three-state longitudinal emergency subsystem and fitted a polynomial controller to its emergency policy with a PAC guarantee on approximation error. Computed stochastic barrier certificates (SBCs) using sum-of-squares (SOS) optimization via semidefinite programming (SDP). Iteratively refined controller coefficients using the expected-decrease condition to raise the certified reach-avoid probability lower bound, then applied the verified controller within the specified state domain.
- **Results:** Compared controllers in closed-loop simulations of normal following, hard braking, and cut-ins. Online constraints reduced headway and road-boundary violation rates, and smoothing reduced command fluctuations. Timed single-step policy inference and QP solves. Computed a reach-avoid probability lower bound for the longitudinal subsystem under polynomial emergency control and assessed its conservatism using Monte Carlo estimates and confidence intervals.

Facial Recognition Attendance System | Undergraduate Capstone Dec 2022 - Jun 2023
Second Prize, University Outstanding Undergraduate Thesis Award. Built a multi-platform attendance system using face detection and recognition on captured or uploaded images, with attendance summaries.

Chongqing Chinasoft International Information Technology Co., Ltd. Jun 2022 - Aug 2022
Internship Project Team Technical Lead & Project Manager | Outstanding Intern; score: 94.
Coordinated **MicFamily** development and integration: karaoke venue management (Spring Boot / MySQL / uni-app).

Honors and Awards

Second Prize, "Huawei Cup" China Post-Graduate Mathematical Contest in Modeling Sep 2024 - Sep 2024
Second Prize (Honourable Mention), 9th "Certificate Authority Cup" Math China International Mathematical Contest in Modeling (CAMCM) Dec 2020 - Dec 2020
Second Prize, 13th and 14th National College Student Mathematics Competition (Chongqing Division) Nov 2021 - Nov 2021

Leadership and Activities

The College, Southwest University | Research Assistant Sep 2023 - Jul 2024
TIPCC, Tsinghua University Press | Appointed Volunteer May 2022 - May 2024
National Higher-Education Expert Committee on Computing and Industry Practice Resource Development
Shapingba District Youth Talent Grassroots Service Program Sep 2021 - Aug 2022
Publicity and Planning Team Lead | University Outstanding Project, 2021 - 2022
The College: Student Union | Academic Affairs & Technology Department Member Sep 2019 - May 2020
Baidu Songguo Elite Program | Outstanding Program Graduate; ranked 2nd by total points Sep 2021 - Dec 2022